



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	SPECTROSCOPY				
Type of Course	Core in Major				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Strong foundation in atomic structure, chemical bonding and electromagnetic radiation and also require knowledge of quantum mechanics.				
Course Summary	The molecular spectroscopy course covers the principles, techniques, and applications of analysing molecular structures and dynamics using various spectroscopic methods.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the principles of molecular spectroscopy	Comprehension	Conceptual	Written exams, quizzes
CO2	Apply spectroscopic techniques to analyse molecules	Application	Procedural	Laboratory reports, projects

CO3	Interpret spectroscopic data accurately	Analysis	Procedural	Problem sets, case studies
CO4	Critically evaluate the limitations of spectroscopic methods	Evaluation	Conceptual	Research papers, presentations
CO5	Demonstrate proficiency in spectral interpretation	Synthesis	Procedural	Oral exams, practical exams
CO6	Relate spectroscopic theory to real-world applications	Application	Conceptual	Research projects, case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	MICROWAVE AND INFRARED SPECTROSCOPY		13	18
	1	The spectrum of non rigid rotator, e.g. of HF, spectrum of symmetric top molecule e.g. of CH ₃ Cl	3	
	2	Instrumentation for Microwave Spectroscopy Stark Modulator, Information derived from Rotational Spectrum	2	
	3	IR Spectroscopy: Born-Oppenheimer approximation	3	
	4	Effect of Breakdown of Born-Oppenheimer approximation	1	
	5	Normal modes and vibration of H ₂ O and CO ₂	2	
	6	Instrumentation for I R Spectroscopy - Fourier transformation I R Spectroscopy	2	
	Sections 6.6, 6.7, 6.8, 6.9, 6.11, 6.13, 6.14, 7.1 – 7.71, 7.12, 7.15, 7.16, 7.17, 7.18 of Book 1			

II	RAMAN SPECTROSCOPY		11	12
	7	Rotational Raman Spectrum of Symmetric top molecules, e.g. of CHCl_3	3	
	8	Combined use of Raman & IR Spectroscopy in structure determination e.g. of CO_2 and NO_3	2	
	9	Instrumentation for Raman Spectroscopy	2	
	10	Non-linear Raman effects, Hyper Raman effect	2	
	11	Stimulated Raman effect and Inverse Raman Effect	2	
	Sections 8.32, 8.4, 8.5, 8.6, 8.7, 8.10, 15.1, 15.2, 15.3, 15.4 of Book1			
III	ELECTRONIC SPECTROSCOPY OF MOLECULES		10	16
	12	Vibrational Analysis of band systems	2	
	13	Deslander's table, Progressions & sequences	2	
	14	Information Derived from vibrational analysis	2	
	15	Franck Condon Principle, Rotational fine structure and P Q and R Branches	2	
	16	Fortrat Diagram, Dissociation Energy, Example of Iodine molecule	2	
	Sections 9.1 – 9.9 of Book 1			
IV	SPIN RESONANCE SPECTROSCOPY		14	24
	17	Interaction of nuclear spin and magnetic field, level population Larmour precession, Resonance Conditions	2	
	18	Bloch equations, Relaxation times, Spin-spin and spin lattice relaxation	3	
	19	The chemical shift, Instrumentation for NMR spectroscopy	2	

	20	Electron Spin Spectroscopy of the unpaired e, Total Hamiltonian, Fine structure, Electron Nucleus coupling and hyperfine spectrum.	3	
	21	Mossbauer Spectroscopy, Resonance fluorescence of γ -rays, Recoilless emission of γ -rays and Mossbauer effect	2	
	22	Chemical shift, effect of magnetic field. Eg. of Fe ₅₇	2	
	Sections 10.1 – 10.9, 11.1 – 11.5.4, 13.1 – 13.5 of Book 1			
V	OPEN ENDED MODULE: ATOMIC SPECTROSCOPY			

Books and References:

1. Molecular structure and Spectroscopy by G. Aruldas (Book 1)
2. Principles of Molecular Spectroscopy by Colin N. Banwell and Elaine M. McCash
3. Spectra of Atoms and Molecules by Peter F. Bernath
4. Molecular Spectroscopy by Jeanne L. McHale
5. Molecular Quantum Mechanics by Peter W. Atkins and Ronald S. Friedman
6. Symmetry and Spectroscopy: An Introduction to Vibrational and Electronic Spectroscopy by Daniel C. Harris and Michael D. Bertolucci

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	0	0	3	1	3	1	1	0	3	1	0
CO 2	3	2	1	0	3	1	3	1	1	1	3	1	0
CO 3	3	2	1	0	3	1	3	1	1	1	3	1	0
CO 4	3	2	2	0	3	1	3	2	1	1	3	1	0
CO 5	3	2	1	0	3	2	3	2	2	1	3	1	0
CO 6	3	1	2	1	3	2	3	2	1	1	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	